The Definition of the Derivative

Section 1: Difference Quotients & Limit Definitions

1 Which limit expression represents the derivative $f'(x)$ by definition?

a. $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

b. $\lim_{h\to 0} \frac{f(x+h)+f(x)}{h}$

c. $\lim_{h\to \infty} \frac{f(x+h)-f(x)}{h}$

d. $\frac{f(x+h)-f(x)}{h}$

2 Evaluate $\lim_{h\to 0} \frac{(3+h)^2-9}{h}$.

a. 3

b. 6

c. 9

d. 0

3 Which function and value of c are represented by $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$?

a. $f(x)=\sqrt{x}$ at $c=4$

b. $f(x)=x^2$ at $c=2$

c. $f(x)=\sqrt{x}$ at $c=2$

d. $f(x)=\frac{1}{\sqrt{x}}$ at $c=4$

Section 2: Differentiability Implies Continuity

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4 If a function f is differentiable at $x = c$, which of the following statements MUST be true?

a. f is continuous at $x = c$

b. $f'(c) > 0$

c. $f''(c)$ exists

d. f has a local extremum at $x = c$

5 Why is the function $f(x) = |x - 3|$ not differentiable at $x = 3$?

a. The left-hand and right-hand derivative limits are unequal (corner)

b. $f(3)$ is undefined

c. f is not continuous at $x = 3$

d. $\lim_{x \to 3} f(x) = \infty$

6 The graph of $f(x) = x^{1/3}$ has what feature at $x = 0$ that prevents differentiability?

a. A vertical tangent line where $f'(x) \to \infty$

b. A jump discontinuity

c. A sharp cusp

d. A hole

Section 3: Tangent and Normal Lines

7 What is the equation of the tangent line to $f(x) = x^3 - 2x$ at the point $(2, 4)$?

a. $y - 4 = 10(x - 2)$

b. $y - 4 = 8(x - 2)$

c. $y - 4 = 12(x - 2)$

d. $y = 10x - 12$

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8 What is the slope of the normal line to $y = x^2 + 3x$ at $x = 1$?

a. $-\frac{1}{5}$

b. 5

c. -5

d. $\frac{1}{5}$

9 At what point on the parabola $y = x^2 - 4x + 3$ is the tangent line horizontal?

a. $(2, -1)$

b. $(2, 0)$

c. $(4, 3)$

d. $(0, 3)$

Section 4: Estimating Derivatives from Tables & Graphs

10 Given values $t = 2, 4, 6$ with $s(2) = 14, s(4) = 30, s(6) = 54$, estimate $s'(4)$ using the average rate of change over $[2, 6]$.

a. 10

b. 8

c. 16

d. 12

11 If the graph of f has a local maximum at $x = 5$ and is differentiable, what is $f'(5)$?

a. 0

b. 1

c. Positive

d. Undefined

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12 If $f'(x) > 0$ for all $x \in (a, b)$, what does this indicate about the graph of f on (a, b)?

a. f is strictly increasing

b. f is strictly decreasing

c. f is concave up

d. f is linear

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